SOLUTIONS CATALOG

Growdea Therapeutic Platforms | 2026

AI-Driven Drug Discovery & Computational Chemistry

Empowering the next generation of therapeutics through intelligent, integrated, and scalable computational platforms

ANALOGUE	ADMET-Vault	iFrag	GrowGen

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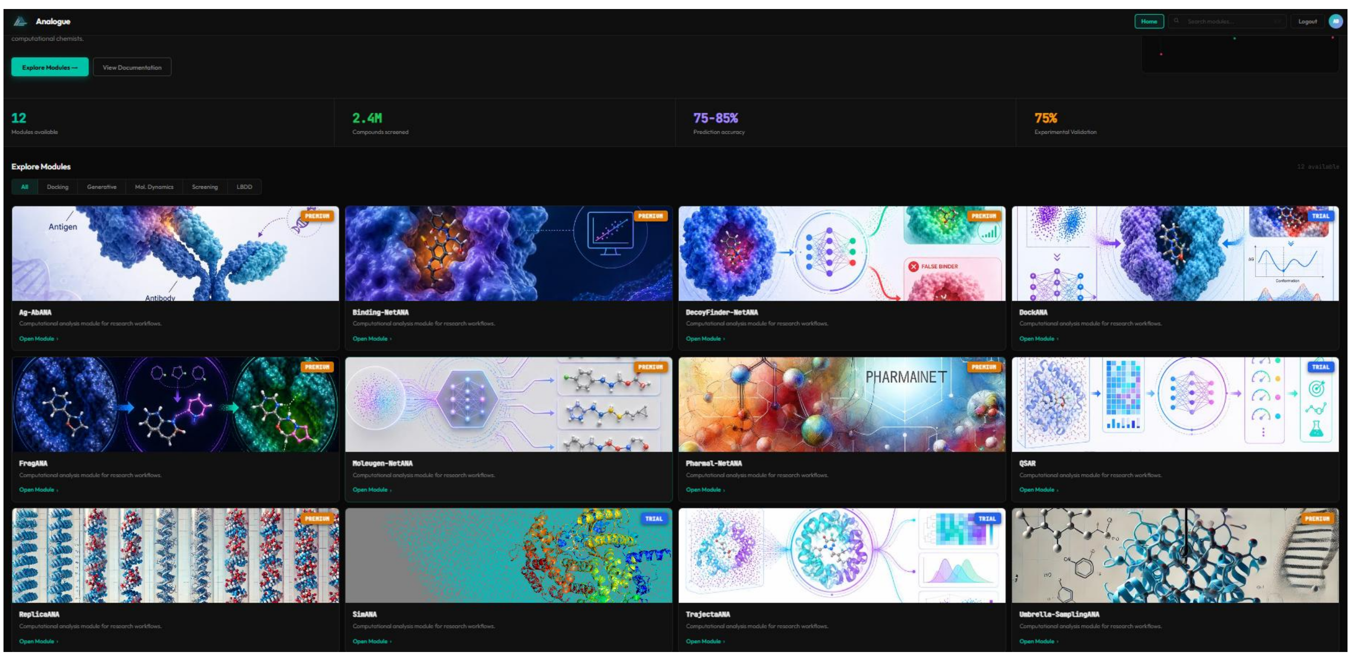
ANALOGUE

12-Module Modular Drug Discovery Suite

ANALOGUE is a comprehensive, AI-powered modular platform purpose-built for pharmaceutical researchers and computational chemists who demand speed, precision, and flexibility. From ligand-based prediction and structure-based screening to molecular dynamics and free-energy calculations, ANALOGUE integrates every critical workflow into a single, cohesive environment — eliminating the friction of switching between disconnected tools.

KEY CAPABILITIES

- QSAR modeling and AI-based protein–ligand binding prediction
- De novo molecule generation trained on 271,914+ bioactive compounds
- High-throughput docking with ML-assisted and physics-based methods
- Full MD simulation pipeline: setup, run, and trajectory analysis
- Enhanced sampling: Umbrella Sampling & Replica Exchange MD (REMD)
- Antigen–antibody interaction analysis and sequence-level optimization



ANALOGUE Platform — Integrated 12-Module Overview

Module Breakdown

01

MoleuGen-NetAna — AI Generative Molecule Design

Rapidly generates chemically diverse, drug-like small molecules using deep generative AI. Trained on 271,914 bioactive compounds from ChEMBL2244 (Kd/IC50/EC50 < 1 μM), it delivers novel scaffolds for target-agnostic lead generation in minutes.

02

Decoy Finder-NetAna — Smart Active/Inactive Classification

Leverages Graph Neural Networks (GNN) to classify compounds as binders or non-binders for a specified protein target. Ideal for dataset curation, virtual screening validation, and model benchmarking.

03

Binding-NetAna — Structural Binding Affinity Prediction

Quantitatively predicts experimental binding affinity from docked protein–ligand complexes using state-of-the-art AI/ML models. Use it as a post-docking confirmation layer to prioritize your highest-affinity candidates.

04

Pharma-AI-NetANA — Sequence-Based Affinity Estimation

Eliminates the dependency on resolved 3D protein structures by predicting binding affinity directly from amino acid sequences and SMILES strings — ideal for novel or difficult-to-crystallize targets.

05

Ag-AbAna — Antibody–Antigen Interaction Optimization

A specialized module for biologics developers. Refine antibody or antigen sequences to enhance binding specificity, stability, and developability — streamlining your biologics lead optimization campaigns.

06

DockAna — Precision Protein–Ligand Docking

Combines ML-assisted and physics-based docking approaches within an intuitive graphical interface, enabling accurate binding pose prediction, interaction fingerprinting, and visual analysis — no command-line required.

07

SimANA — Full-Pipeline Molecular Dynamics

Provides a complete, GUI-driven MD simulation pipeline covering system setup, force field assignment, energy minimization, equilibration, and production runs — making rigorous MD accessible to non-specialists.

08

TrajectaAna — Advanced Trajectory Analysis

An interactive analytics toolkit for MD trajectories: RMSD, RMSF, hydrogen bond occupancy, radius of gyration, conformational clustering, and more — visualized through dynamic, publication-ready plots.

09

FragAna — Fragment-Based Analogue Design

Accelerates lead optimization by systematically replacing scaffold fragments using a library of 700,000+ fragments. Each proposed analogue is coupled with predicted binding scores, enabling rapid SAR exploration.

10

Umbrella Sampling — PMF & Energy Barrier Profiling

Implements enhanced sampling to overcome high energy barriers and access rarely visited conformational states. Calculate the potential of mean force (PMF) with statistical rigor for mechanistic insights.

11

REMD — Replica Exchange Molecular Dynamics

Runs multiple parallel simulations across temperature or Hamiltonian replicas, facilitating periodic exchange to dramatically improve conformational sampling, overcome local minima, and enhance thermodynamic predictions.

12

QSARAna — No-Code QSAR Model Builder

A fully no-code ML environment for QSAR: compute 1,500+ chemical descriptors from SMILES, perform feature selection, train diverse models, and predict activity/property for new compounds — all through an intuitive interface.

ADMET-Vault

AI-Powered In Silico Pharmacokinetic & Toxicity Profiling

Bringing a drug candidate to the clinic requires confident command of its ADMET profile — and ADMET-Vault delivers exactly that. This AI-powered platform predicts Absorption, Distribution, Metabolism, Excretion, and Toxicity parameters with high accuracy, benchmarked against FDA-approved drug distributions. More critically, it goes beyond prediction: ADMET-Vault actively guides lead optimization, suggesting structural modifications to bring off-target parameters back within acceptable ranges.

KEY PREDICTIVE PARAMETERS

Absorption

- Caco-2 cell effective permeability (log Papp, cm/s) — flags high vs. low intestinal permeability
- Human intestinal absorption (binary) — predicts oral uptake potential
- Aqueous solubility (log mol/L) — estimates dissolution risk at physiological conditions
- Blood-brain barrier penetration (binary) — screens out CNS-active liabilities early
- PAMPA permeability (binary) — passive transcellular permeability assessment

Distribution

- Plasma protein binding rate (%) — quantifies free drug fraction available for activity
- Volume of distribution at steady state (L/kg) — predicts tissue vs. plasma partitioning
- Lipophilicity (log-ratio) — flags optimal vs. excessive lipophilicity for drug-likeness

Metabolism

- CYP450 inhibition profiling: CYP1A2, CYP2C9, CYP2C19 — identifies metabolic interference risk
- CYP2D6 and CYP3A4 substrate prediction — maps primary metabolic pathways
- P-glycoprotein inhibition — predicts efflux transporter interactions affecting bioavailability

Excretion

- Hepatocyte clearance ($\mu L \cdot min^{-1} \cdot 10^6$ $cells^{-1}$) — liver-based metabolic clearance rate
- Microsomal clearance ($mL \cdot min^{-1} \cdot g^{-1}$) — in vitro metabolic stability surrogate
- Half-life via Obach model (hr) — estimated systemic exposure duration

Toxicity

- Acute toxicity LD50 (log mg/kg) — systemic acute toxicity risk classification
- AMES mutagenicity (binary) — genotoxicity flag for early safety triage
- Drug-induced liver injury (binary) — hepatotoxicity liability screening

Case Study: Lead Optimization in Action

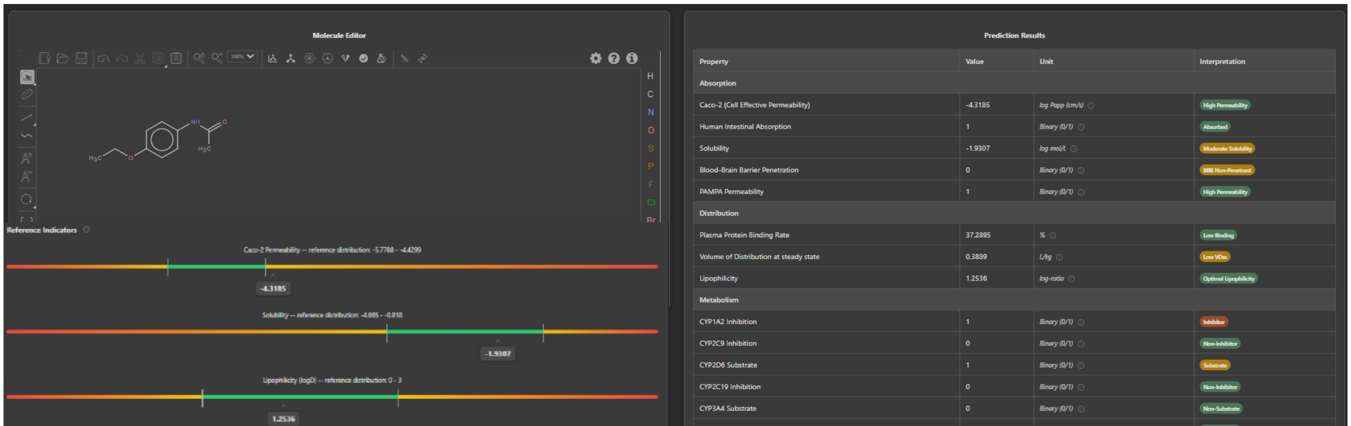
ADMET-Vault goes beyond prediction. When a molecule fails ADMET screening, the platform's built-in Genetic Algorithm (GA) engine automatically searches chemical space for a structurally optimized replacement — turning a critical liability into a viable drug candidate in a single workflow.

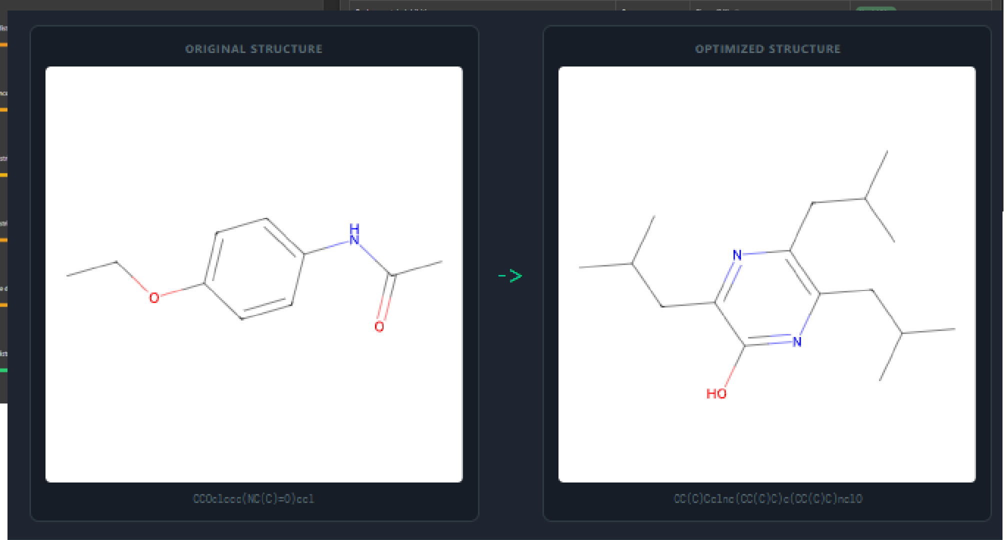
Demonstrated in Practice: Scaffold Hop Case Study

The original molecule (CCOc1ccc(NC(C)=O)cc1) entered optimization with an ADMET score of 0.000 and five failing properties — classified as **Critical Risk**. The acetanilide scaffold was flagged as a known DILI liability, triggering a Stage 2 scaffold hop. The GA-evolved candidate achieved an **ADMET score of 0.8155 (+0.8155 improvement)**, resolving three of five issues:

- **AMES Mutagenicity** — Mutagenic → Non-mutagenic
- **Drug-Induced Liver Injury (DILI)** — High risk → Low risk
- **CYP2D6 Substrate** — Substrate → Non-substrate (more predictable metabolism)
- **Volume of Distribution** — Improved to 1.694 L/kg (balanced tissue distribution)
- **Acute Oral Toxicity (LD50)** — Improved to 3.801 log mg/kg

Both the original and optimised molecules maintained full **Lipinski Rule-of-Five compliance** (zero violations), confirming drug-likeness is preserved through optimization.





ORIGINAL VS OPTIMISED -- ALL 12 PROPERTIES

Property	Original	Optimised	Change
Caco-2 Permeability	-4.319	-4.583	-0.265
Human Intestinal Absorption	[OK] Absorbed	[OK] Absorbed	—
Aqueous Solubility	-1.931	-3.999	-2.068
BBB Penetration	[OK] Non-Penetrant	[OK] Non-Penetrant	—
PAMPA Permeability	[OK] High Permeability	[OK] High Permeability	—
Plasma Protein Binding	37.289	68.248	+30.959
Volume of Distribution	0.389	1.694	+1.305
Lipophilicity (logD)	1.254	2.775	+1.522
CYP1A2 Inhibition	[!] Inhibitor	[OK] Non-Inhibitor	—
CYP2C9 Inhibition	[OK] Non-Inhibitor	[OK] Non-Inhibitor	—
CYP2C19 Inhibition	[OK] Non-Inhibitor	[OK] Non-Inhibitor	—
CYP2D6 Substrate	[!] Substrate	[OK] Non-Substrate	—
CYP3A4 Substrate	[OK] Non-Substrate	[!] Substrate	—
P-gp Inhibition	[OK] Non-Inhibitor	[OK] Non-Inhibitor	—
Hepatocyte Clearance	27.769	21.914	-5.855
Microsomal Clearance	18.458	18.631	+0.173
Half-Life (Obach)	2.942	6.260	+3.318
Acute Toxicity (LD50)	3.295	3.801	+0.506
AMES Mutagenicity	[!] Mutagenic	[OK] Non-mutagenic	—
Drug-Induced Liver Injury	[!] Toxic	[OK] Non-toxic	—

Key Capability Highlights

- Automated scaffold hops when Stage 1 optimization ceiling is detected
- GA-driven exploration with per-candidate ADMET scoring and Tanimoto tracking
- AI-assisted medicinal chemistry analysis with property-specific fix strategies
- Full PDF report generated per optimization run — audit-ready and

iFrag

Interaction-Guided Fragment Analysis & Replacement Platform

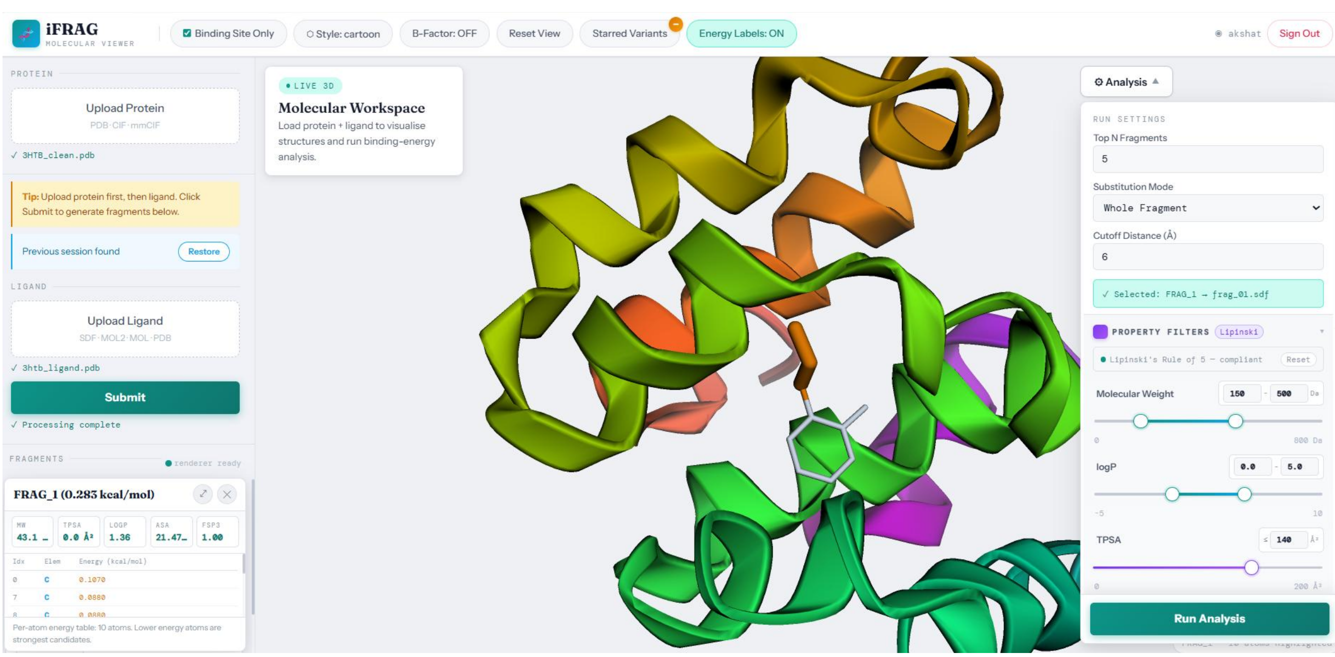
iFrag transforms the way medicinal chemists approach lead optimization. By analyzing docked protein–ligand complexes at the fragment level using MM/GBSA energy decomposition, iFrag pinpoints the exact molecular fragments responsible for unfavorable interactions — then intelligently proposes validated replacements from a curated library of 708,687 fragments, guided by the specific interaction patterns of the local protein environment.

HOW IFRAG WORKS

- Automated energy minimization removes steric clashes and optimizes geometry
- MM/GBSA per-atom energy decomposition identifies high-energy (weak) fragments
- Molecular fragmentation preserves chemically meaningful substructures and atom indices
- Residue environment analysis (within 6.5 Å) classifies nearby protein residues by interaction type
- 708,687-fragment library screened for best-fit replacements matching required interaction patterns

Why iFrag Delivers Better Lead Candidates

Traditional analogue design relies on chemical intuition. iFrag replaces guesswork with quantitative, structure-informed guidance — ensuring every proposed modification is evaluated against the exact binding environment of your target. The result: higher-quality analogues, shorter optimization cycles, and improved success rates in subsequent docking and MD validation.



GrowGen

Transformer-Based De Novo Molecule Generation

GrowGen is Growdea's flagship generative AI platform for exploring drug-like chemical space. Built on transformer-based autoregressive modeling with Rotary Position Embedding (RoPE) and property-conditioned reinforcement learning, GrowGen produces novel, valid, and highly drug-like molecules at scale — and goes one step further by enabling direct docking of generated candidates against your protein target of interest.

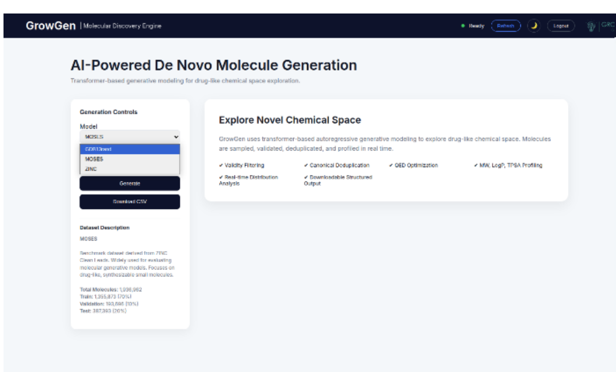
GENERATION & VALIDATION PIPELINE

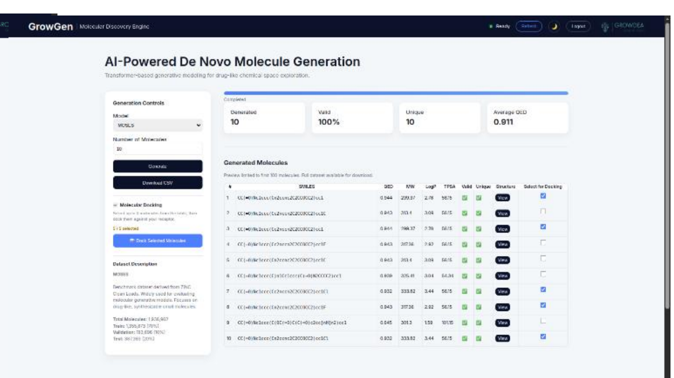
- Transformer autoregressive model with RoPE and descriptor embedding
- Policy-gradient reinforcement learning optimizing molecular validity, global uniqueness, and QED
- Real-time validity filtering and canonical SMILES deduplication
- MW, LogP, TPSA, and QED profiling with live distribution analysis
- One-click docking of selected generated molecules against target protein
- Downloadable, structured output for downstream screening or synthesis

BENCHMARK PERFORMANCE

Dataset	Validity	Uniqueness	Novelty	Avg QED
GDB13rand	>82%	>91%	>99%	Improved
MOSES	>85%	>93%	>99%	Improved
ZINC	>80%	>90%	>99%	Improved

RL-enhanced generation consistently exceeds 80% chemical validity, 90% uniqueness, and 99% novelty across all benchmark datasets — delivering a generative engine you can trust for real drug discovery campaigns.





Generate selected number of molecules with high QED score. The tool also allows use to dock the novel generated molecule against the desired protein drug target. Any molecule that are generated could be selected by user and tested against the protein.

Ready to Accelerate Your Drug Discovery?

Our team of computational chemists and AI researchers is ready to support your pipeline — from early-stage target identification to lead optimization and candidate selection.

Get in touch to schedule a platform demonstration.

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